<u>Element A</u>	<u>Information</u> **Element A: Presentation and Justification of the Problem**
Problem Statement and Details (A1, A2, A5) **Research**	***Explain your problem in detail, including details that you might consider common knowledge. Use data and statistics to justify your problem statement. Do not provide a bulleted list of facts; rather, organize the problem details in paragraph form. Address as many of the 5 Ws +1 H (who, what, where, when, why, how) as is possible, for example:*** ***1) Who is involved in this problem?*** - *stakeholders such as Dr Jackson (the client), the engineers (our group)* ***2) What are they doing?*** - *The client will be stating his problem to the engineer for them to find/build a solution to the problem. The engineer will then receive feedback from the client about the solution to the problem and the engineer will revise and apply the new ideas/additions to the build.* ***3) What unique situations does the problem occur within, if any?*** - *The situation is an outside 12x12in garden that needs to be transportable, use fresh water collection, and supply water for a period of 10 days without the client's attention. It has to be able to supply the plants with the appropriate amount of water for their type and season.* ***4) What is the size of the issue?*** - *The garden is 12x12 inches with a 3-4ft distance from the downspout coming from the roof of the house.* ***5) Where is the problem happening?*** - *The problem is happening at the client's house, in his backyard garden.* ***6) When does the problem occur?*** - *The client will be leaving to go on trips during the summer time and will be away from his house for a maximum of 10 days. The plants in the garden will need to be watered while the client is gone using fresh collected water.* ***7) Why does the problem happen?*** - *The client wants it to use fresh collected water to lower the need of the municipal water center. If the plants are not watered while the client is gone then they will die, especially in the summer months.* ***8) How often does the problem happen?*** - *The problem happens when the client leaves his home and has no one to water his plants and can not rely on it raining everyday. There needs to be a storage of the collected water so that there is a sufficient amount of water given to the plants*

	everyday. ***In order to earn a score of 5, you will need to cite at least five sources that are from the last ten years and that come from 3 or more of the following credible categories: peer reviewed journals, professional organizations, government agencies, local and national newspapers. Use in-line citations to explicitly state which information is from which source. Do not just list all of the citations at the end of the submission or in footnotes.*** https://www.gardendesign.com/how-to/summer-watering.html ***In my research I have found that watering in the morning is the best time to water because it allows the plants to absorb the water before the heat evaporates during the day. It is also best to water the base of the plants and not the leaves or vegetation to prevent disease. Lastly, not letting the soil get too dry in the summer months allows for the plants to grow and maintain well draining soil with nutrients.*** https://weatherspark.com/countries/s/US/VA/2 - **Summer July - August in Virginia Ranges from 75-88 F typically** - **Fall August - November in Virginia Ranges from 43-77 F typically** https://www.freshwatersystems.com/blogs/blog/5-advantages-and-disadvantages-of-rainwater-harvesting?srsltid=AfmBOoq9Nl4VUHc9zO7lBMeNPw-rJYmRJJ9FcuQtDh_Qu5aYWY9oKFa5 **Pros and Cons of the Collection of Rainwater: Good for the environment and the surrounding soil that it is distributed to. The collection of rainwater prevents erosion. Reduces the amount of energy used because it is not from the municipal center. Low maintenance and low costs. Allows for the storage of water in times of drought and the absence of gardeners. Better water quality than city water that has chemicals. Unpredictable collection amounts. Can become contaminated with algae or fungus if not taken care of. Takes up space.** https://extension.usu.edu/yardandgarden. **Plants such as Squash, beans, and tomatoes typically need 1 to 2 inches of irrigation per week.** https://cropsandsoils.extension.wisc.edu/articles/the-important-role-of-soil-texture-on-water/#:~:text=Soil%20texture's%20influence%20on%20water,rates%20than%20fine%20textured%20soils **Water infiltration speed and plant water availability Sandy soils with bigger pore spaces. Increase water movement and have higher infiltration. Coarse soils and soak up rain, but are unable to hold as much water as finer solids. Sandy soils have a low water holding capacity compared to loamy or clayey soils. Soils with clay contents hold water tightly and provide less**

	water storage for the loamy soils.
Stakeholder Concerns and Connections (A3, A4)	***Directly indicate at least five primary stakeholders or stakeholder groups. These are the people most directly impacted by the problem. Highlight the concerns of your primary stakeholders and use that to justify why there is a need to address the problem. Consider the breadth of the problem and how it might have social and environmental impacts on secondary stakeholders, which are people who may be impacted by a solution to the problem and are still part of the system.***
Primary Stakeholders and concerns **Secondary Stakeholders and concerns**	Primary Stakeholders 1. Dr. Jackson - Needs an irrigation system that runs off of a collection system that lasts for a max of 10 days during the summer months while he leaves. Need to be transportable. 2. Engineering group (Gabby D., Tez, Gabby S. Ethan R.) - Building the product and making a solution for Dr. Jackson issue/concern. Revising and editing products for Dr. Jackson's feedback. 3. Investors - Want to build the product and market the product in a wide range. Invest to make the product better quality and have mass production to have larger sales and larger profit. 4. Amazon/retailer where we buy materials from - Sells us the materials that we need to be able to build the product. Make money off of the units that we sell because we need materials for every unit. 5. Other customers - Other gardeners that own a business that would like to market the product in their stores. See housing the product in their stores would be beneficial for their sale and their profits/expansion of their business. Secondary Stakeholders 1. Municipal water centers - Able to conserve water because of the gardeners using natural water for their gardens. 2. Impacts the environment - Allows for the beneficial use of resources, rainwater, in homes. Allows for natural processes such as nutrients and minerals in the rain water to be distributed to plants instead of having chemicals from the municipal center.

Problem Importance (A6)	***End Element A with a compelling argument for why this problem is important and needs to be solved now. Support this argument with data and statistics.*** This problem is important because many civilians grow a garden during the summer and have taken up an interest in growing their own food. Civilians also still plan to vacation but need someone to care for their garden especially during the hotter and dryer months such as summer. Much of the human population runs their water off of the municipal center which gets expensive when using it for outside uses such as the garden when drought occurs. To solve this problem an irrigation system that works off of a collection of fresh water and is able to store water for extended periods while there is a drought or the gardener is away is needed. Civilians are able to save approximately $35 dollars a month according to *House Logic* when using a collection method for watering their garden. According to *Travel Agency Central* 85% of Americans travel during the summer months and about 55% of Americans have some sort of at home garden, according to *Raleigh Realty,* so there will be many customers that are interested in the product because it will take care of their plants while they are away while also saving money. https://www.houselogic.com/save-money-add-value/save-on-utilities/water-savings-barrel/ https://www.travelagentcentral.com/your-business/stats-85-percent-americans-taking-least-one-summer-vacation https://raleighrealty.com/blog/gardening-statistics-trends